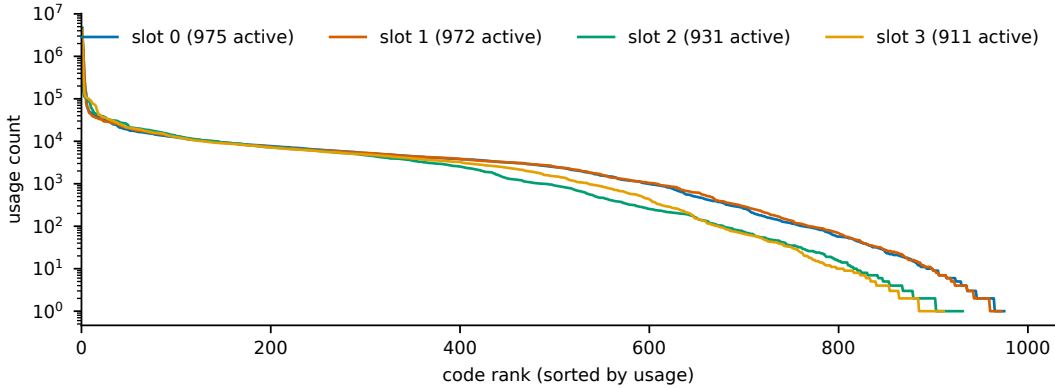


Supplementary Information

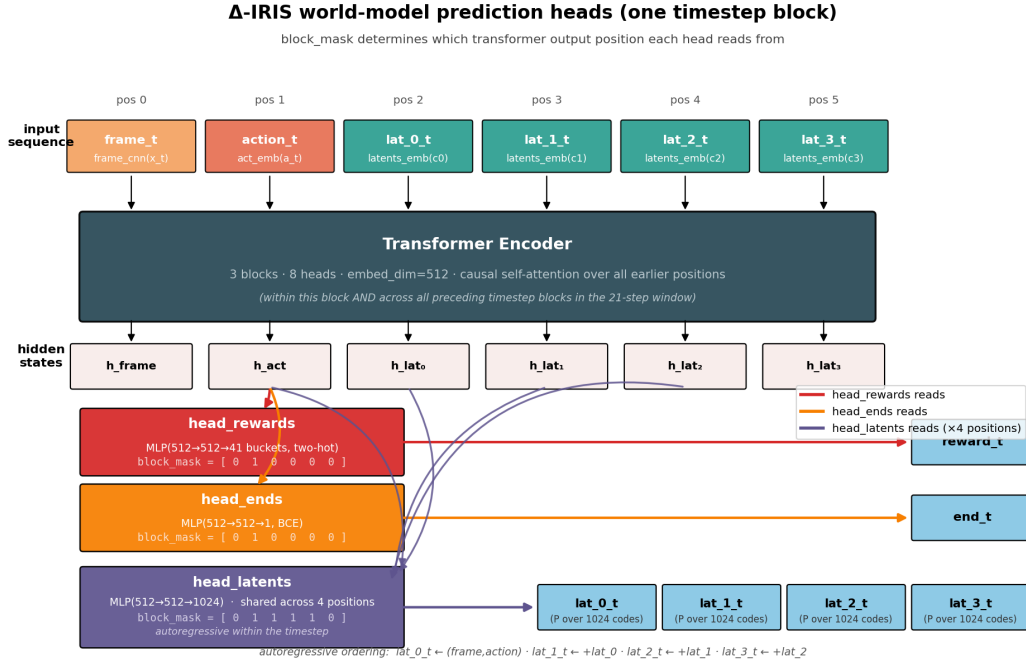
From Codes to Causes: A Causal Interpretability Audit of Quantized Delta Representations in a Transformer World Model

R. Xia

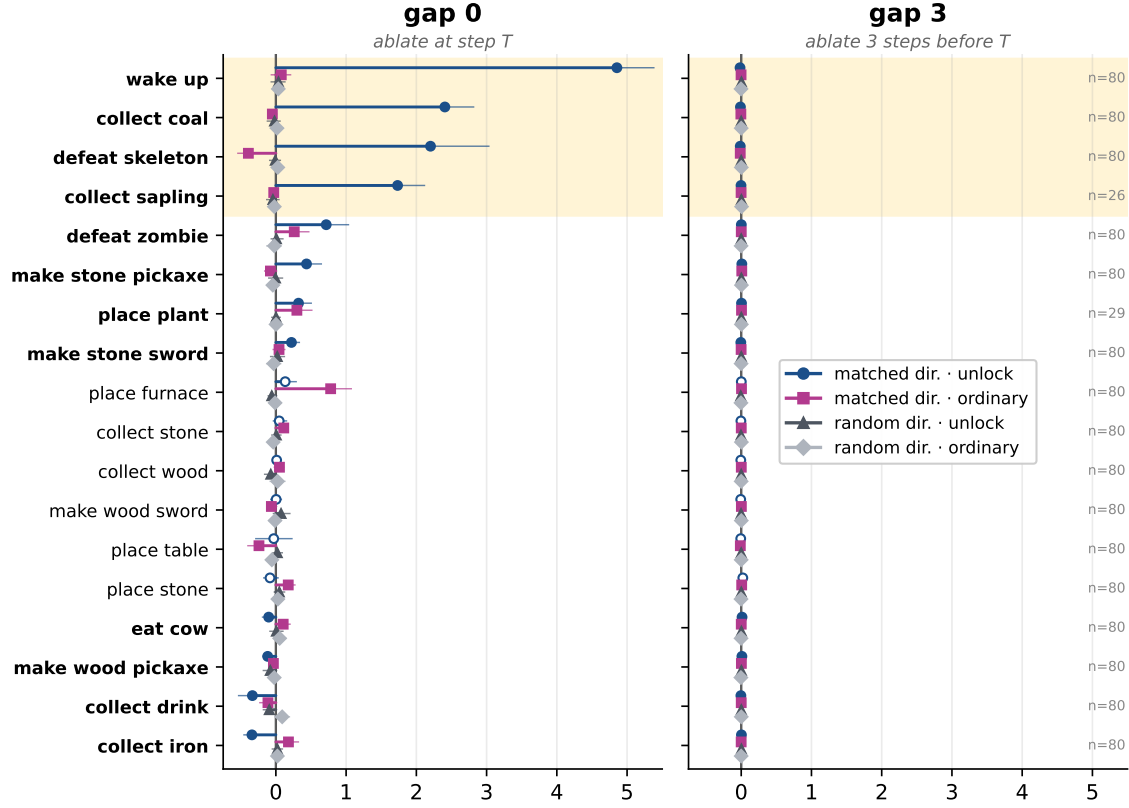
This document collects the displays deferred from the main text: codebook utilisation (Fig. S1), the prediction-head read positions (Fig. S2), single-step ablation (Fig. S3), decoded-frame filmstrips (Fig. S4) and the partially observed (no-HUD) control (Fig. S5); and Tables S1–S8: detector codes, cross-corroborated dictionary features, matched-protocol reconstruction, HUD-shortcut margins, the sparse-autoencoder robustness sweep, the empirical random-direction null, steering, and direct logit attribution. All re-use the single frozen agent and the saved rollout corpora described in Methods.



Supplementary Figure S1: **Codebook utilisation over the full 10^7 -transition replay buffer.** Per-slot code-usage counts, sorted descending (log scale). All 1024 codes are active in the union of slots, but usage is Zipf-like: the head codes encode “no change in this quadrant” and the event vocabulary occupies the tail.

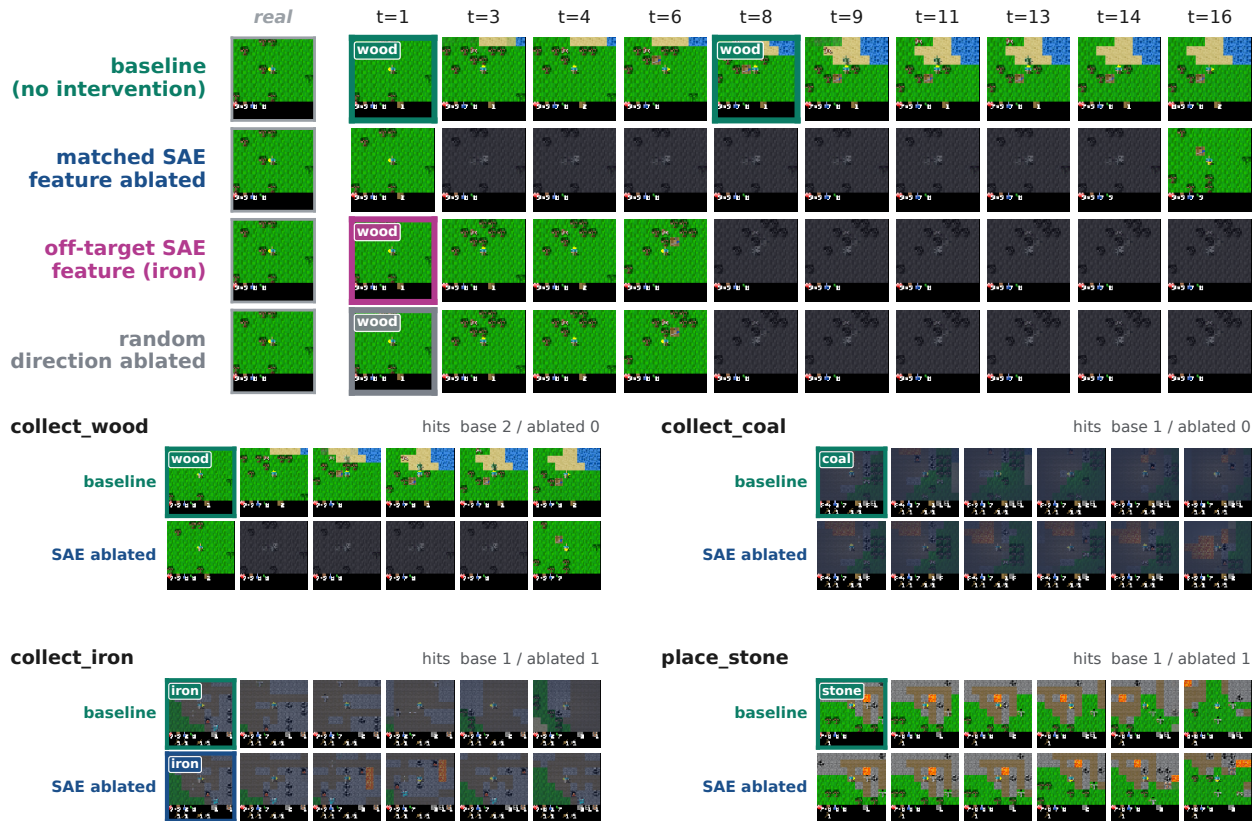


Supplementary Figure S2: **Detailed view of the Δ -IRIS prediction heads** (architecture follows Micheli et al.; our redrawing). **head_rewards** and **head_ends** read the post-transformer activation at the action position (block mask [0, 1, 0, 0, 0, 0]); **head_latents** reads positions 1–4 (mask [0, 1, 1, 1, 1, 0]) and predicts the four latent codes autoregressively within the timestep. The final latent position feeds subsequent attention but carries no prediction head.

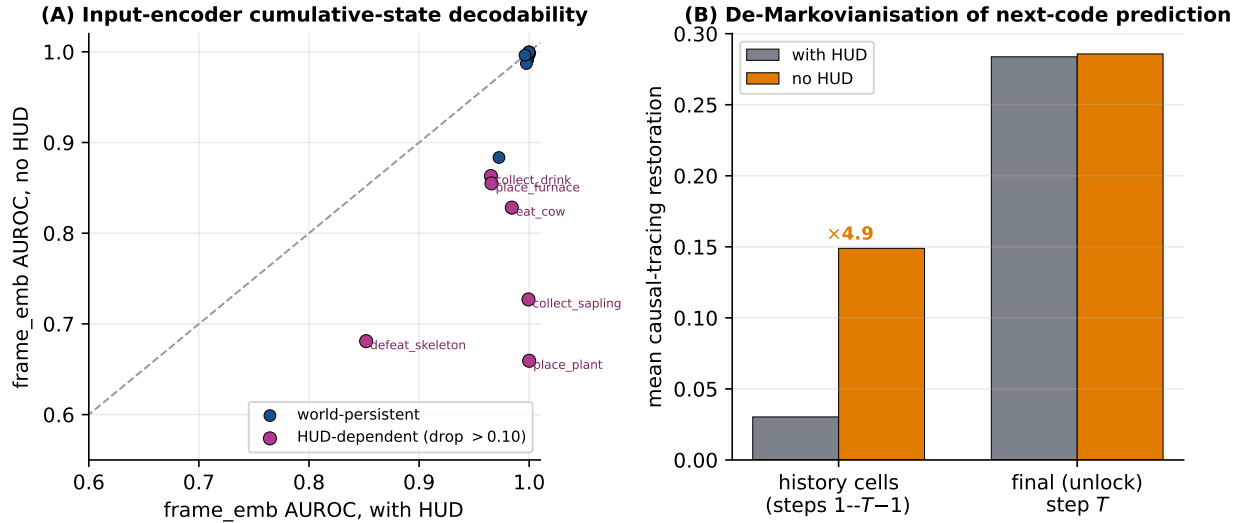


$\Delta \log P(\text{realised codes at } T) = \text{baseline} - \text{ablated}$ (positive $\Rightarrow$ ablation lowers the likelihood of the realised codes)

Supplementary Figure S3: **Single-step causal ablation.** $\Delta \log P = \text{baseline} - \text{ablated}$ log-likelihood of the four realised codes at the target step (positive $\Rightarrow$ ablation lowered their likelihood) when a direction is projected out of the block-1 residual stream, at the target step (gap 0, left) or three steps earlier (gap 3, right), on one shared horizontal scale. Markers are means; whiskers 95% bootstrap CIs; the full 2×2 of {matched, random} direction $\times$ {unlock, ordinary} moment is shown. Random directions are inert; the matched direction is selective to unlock moments only for the dominant concepts; effects vanish entirely at gap 3 (the interventional face of the HUD shortcut).



Supplementary Figure S4: **Imagined trajectories under ablation (decoded frames)**. Top: paired trajectories for **collect_wood** from a shared burn-in and sampling seed—no intervention; matched SAE feature ablated (wood never appears, dream degrades); off-target SAE feature ablated (the **collect_iron** feature, a magnitude-comparable control); random unit direction. Only the matched direction erases the event. Bottom: baseline vs matched-SAE ablation for four concepts, ranging from clean erasure (**collect_wood**) through partial suppression (**collect_coal**) to none (**collect_iron**, **place_stone**)—honest negatives whose best SAE features are not causally load-bearing.



Supplementary Figure S5: **Partially observed (no-HUD) control: with-HUD vs no-HUD.**

A second agent (identical code, hyperparameters and seed) was trained on a Crafter variant that zeroes the on-screen inventory/vitals region; mean return falls from 16.2 to 9.7. **(A)** input-encoder (frame_emb) held-out AUROC for each cumulative-state concept, with HUD (x) vs no-HUD (y). World-persistent achievements (blue; placed objects, gathered materials, tools) stay near 1.0 because they leave visible marks in the scene, so the input-encoder shortcut survives; only the six genuinely HUD-only concepts (magenta) drop below the diagonal. A transformer block out-reads frame_emb on only 1 of 17 concepts, so the hidden state is not made linearly decodable from the blocks. **(B)** causal-tracing restoration mass on history cells (steps 1 to $T-1$) vs the final unlock step: with state hidden, mean historical restoration rises $\approx 5\times$ ($0.03 \rightarrow 0.15$) while the final-step mass is unchanged—next-code prediction de-Markovianises.

Supplementary Table S1: **Strongest detector code per achievement** (selection). n_{co} : co-occurrences of the code with the just-unlocked event; n_c : total occurrences; lift = $P(u | c)/P(u)$. All p -values are below 10^{-40} (uncorrected exact binomial); the full 431-cell table ships with the code release.

Achievement	Code	n_{co}/n_c	$P(u c)$	Lift
place_plant	(s_3, c_{938})	477/519	0.92	543×
collect_sapling	(s_2, c_{707})	419/463	0.90	535×
collect_iron	(s_3, c_{75})	360/410	0.88	708×
collect_coal	(s_3, c_{578})	450/626	0.72	467×
make_wood_pickaxe	(s_3, c_{903})	19/35	0.54	325×
place_table	(s_3, c_{36})	480/1507	0.32	188×
eat_cow	(s_2, c_{648})	251/1127	0.22	134×
place_furnace	(s_0, c_{51})	89/412	0.22	132×
collect_wood	(s_3, c_{940})	474/2471	0.19	113×
defeat_zombie	(s_1, c_{727})	68/365	0.19	113×

Supplementary Table S2: **Cross-corroborated SAE features**. The feature’s top achievement (lift within its top activation quartile), most-aligned CAV, and the (slot, code) with the highest conditional mean activation. The first five satisfy strict triple-corroboration ($|\cos| > 0.4$ to the same-named CAV); f_{372} is the third member of the (s_3, c_{36}) split.

Feature	Achievement (lift)	CAV (cos)	Code
f_{1742}	collect_coal (589×	just[collect_coal] (+0.58)	(s_3, c_{578})
f_{1820}	collect_iron (624×	just[collect_iron] (+0.41)	(s_3, c_{75})
f_{872}	make_stone_pickaxe (522×	just[make_stone_pickaxe] (+0.48)	(s_3, c_{616})
f_{677}	place_table (527×	just[place_table] (+0.45)	(s_3, c_{36})
f_{1451}	make_wood_pickaxe (548×	just[make_wood_pickaxe] (+0.46)	(s_3, c_{36})
f_{372}	make_wood_sword (444×	just[make_wood_sword] (+0.23)	(s_3, c_{36})

Supplementary Table S3: **Matched-protocol reconstruction** on a common 8,192-transition held-out batch. Perplexity is $\exp(\text{entropy})$ of the token histogram over all slots. Δ -IRIS conditions its decoder on (x_t, a_t) ; the baselines reconstruct x_{t+1} from codes alone, so Δ -IRIS’s lower error is expected and not a like-for-like comparison.

Tokenizer	MSE ↓	PSNR ↑	LPIPS ↓	perplexity	tokens/frame
Δ -IRIS (delta + action)	0.00021	36.7	0.048	80.7	4
frame-only ablation	0.0145	18.4	0.467	48.5	4
original IRIS	0.0056	22.5	0.080	109.2	16

Supplementary Table S4: **HUD-shortcut margins with bootstrap CIs**. For each cumulative-state concept, `frame_emb` held-out AUROC and the *worst* (smallest) AUROC difference against any transformer block, with a paired-bootstrap 95% CI ($B=1000$). `frame_emb` is not significantly below any block on 16 of 18 concepts; the two exceptions (`collect_sapling`, `place_plant`) trail by $< 10^{-3}$, with all representations above 0.999.

Concept	<code>frame_emb</code> AUROC	worst Δ AUROC	95% CI	not below
<code>collect_coal</code>	1.000	+0.046	[+0.043, +0.049]	yes
<code>collect_drink</code>	0.965	+0.046	[+0.040, +0.051]	yes
<code>collect_iron</code>	1.000	+0.065	[+0.061, +0.068]	yes
<code>collect_sapling</code>	0.999	−0.0007	[−0.0010, −0.0004]	no
<code>collect_stone</code>	0.999	+0.014	[+0.013, +0.016]	yes
<code>collect_wood</code>	1.000	+0.0015	[+0.0010, +0.0020]	yes
<code>defeat_skeleton</code>	0.852	+0.009	[+0.004, +0.013]	yes
<code>defeat_zombie</code>	0.973	+0.019	[+0.016, +0.022]	yes
<code>eat_cow</code>	0.984	+0.012	[+0.009, +0.015]	yes
<code>make_stone_pickaxe</code>	1.000	+0.018	[+0.016, +0.019]	yes
<code>make_stone_sword</code>	1.000	+0.020	[+0.018, +0.022]	yes
<code>make_wood_pickaxe</code>	1.000	+0.008	[+0.007, +0.009]	yes
<code>make_wood_sword</code>	1.000	+0.042	[+0.039, +0.046]	yes
<code>place_furnace</code>	0.966	+0.015	[+0.012, +0.019]	yes
<code>place_plant</code>	1.000	−0.0000	[−0.0000, −0.0000]	no
<code>place_stone</code>	0.997	+0.017	[+0.015, +0.018]	yes
<code>place_table</code>	0.996	+0.016	[+0.013, +0.018]	yes
<code>wake_up</code>	0.916	+0.034	[+0.029, +0.041]	yes

Supplementary Table S5: **Sparse-autoencoder robustness sweep** on the block-1 summary activations (81,919 samples, 60 epochs), and single SAEs at other layers. FVU: fraction of variance unexplained on episode-disjoint validation. Detector features reappear as each variant’s own top detector (mean matched decoder cosine 0.63).

Axis	Config (M , k , seed / layer)	FVU	dead features
width	1024, 16, 0	0.0445	0
(default)	2048, 16, 0	0.0440	0
width	4096, 16, 0	0.0441	0
sparsity	2048, 8, 0	0.0543	28
sparsity	2048, 32, 0	0.0363	0
seed	2048, 16, 1	0.0439	0
layer	frame_emb	0.267	—
layer	wm_input	0.0475	—
layer	block_2	0.0265	—
layer	block_3	0.214	—

Supplementary Table S6: **Empirical random-direction null**. Single-step matched-SAE ablation effect (gap 0) against an $R=50$ random-direction null; $z = (\text{matched} - \mu_{\text{null}})/\sigma_{\text{null}}$. The matched effect clears the 99th percentile for three of eight dominant concepts; honest negatives (e.g. `collect_iron`) sit at or below the null.

Concept	matched $\Delta \log P$	null $\mu \pm \sigma$	z	$> p_{99}$
<code>collect_coal</code>	+2.70	-0.02 ± 0.11	24.2	yes
<code>collect_sapling</code>	+1.73	$+0.02 \pm 0.17$	9.9	yes
<code>make_stone_pickaxe</code>	+0.61	-0.01 ± 0.20	3.1	yes
<code>place_plant</code>	+0.32	$+0.02 \pm 0.21$	1.4	no
<code>make_wood_pickaxe</code>	-0.16	-0.01 ± 0.15	-1.0	no
<code>place_table</code>	-0.04	-0.04 ± 0.20	0.0	no
<code>place_stone</code>	-0.11	-0.03 ± 0.19	-0.5	no
<code>collect_iron</code>	-0.39	-0.02 ± 0.16	-2.3	no

Supplementary Table S7: **Steering (sufficiency) inside imagination.** Adding the matched SAE direction ($h += \alpha \hat{d}$) from ordinary moments raises within-horizon detector-code probability only weakly; suppression (matched ablation) is shown for comparison. Mean induction gain +0.04 vs suppression drop +0.07 across 12 concepts (mean activate–suppress = -0.03): no activate-over-suppress asymmetry in this agent.

Concept	induce base	induce gain	suppress base	suppress drop
collect_sapling	0.00	+0.25	0.60	+0.20
collect_stone	0.15	+0.20	1.00	+0.00
collect_iron	0.05	+0.05	1.00	+0.15
make_stone_pickaxe	0.10	+0.05	0.80	+0.10
place_stone	0.05	+0.05	1.00	+0.00
collect_coal	0.05	−0.05	0.95	+0.25
eat_cow	0.15	−0.05	0.75	+0.00
make_stone_sword	0.10	+0.00	0.75	+0.30
wake_up	0.00	+0.00	0.30	−0.20

Supplementary Table S8: **Direct logit attribution** to the detector code, mean over 18 concepts ($n=40$ unlock moments each). $|\overline{\text{det}}|$: mean absolute attribution to the detector code; $|\overline{\text{det}}| - |\overline{\text{ctrl}}|$: detector-vs-control specificity; #sig: concepts whose detector attribution CI excludes 0. The SAE direction’s attribution is $\sim 4\times$ the CAV’s and $\sim 24\times$ a random direction’s; the sign is concept-dependent (off-path `collect_iron` -1.56 , `wake_up` $+1.50$).

Direction	$ \overline{\text{det}} $	$ \overline{\text{det}} - \overline{\text{ctrl}} $	#sig / 18
SAE feature	0.311	+0.073	9
CAV	0.083	+0.024	6
random	0.013	+0.000	2